Sharp dimension bounds for right-angled hyperbolic polytopes

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Abstract

We prove that a finite-volume right-angled hyperbolic polytope has dimension at most eight, and at most four if all its vertices are ideal. Both bounds are attained. The proofs share an elementary projection: from a finite interior point, project to a facet of maximum unit-normal height. Repeating this operation three times in dimension nine gives a six-face whose wall directions form a spherical code with pairwise products strictly below $1/4$. A cubic tensor-square identity bounds this code by 26, contradicting the published minimum of 27 facets. For the ideal bound, one projection in dimension five combines Musin's kissing-number theorem and Kolpakov's 24-cell rigidity; an opposite-normal calculation excludes the metric equality case.

1 Introduction

Right-angled hyperbolic polytopes provide fundamental domains for reflection groups, but their possible dimensions have remained separated from the dimensions of the known constructions. Potyagailo and Vinberg [5] construct finite-volume examples through dimension eight. Dufour [2] proves that their dimension is at most twelve. For polytopes whose vertices are all ideal, the regular right-angled 24-cell gives an example in dimension four, whereas Kolpakov [3, Corollary 7] obtains the upper bound six. We close these two gaps. The compact case already has the sharp dimension bound four; see [5, Section 2].

A polytope is *entirely ideal* if all its vertices are ideal. Throughout, hyperbolic polytopes have finitely many facets and finite volume, and the curvature is normalized to -1. A right-angled polytope has dihedral angle $\pi/2$ at every actual ridge. Finite and ideal vertices are otherwise allowed in any combination, and no arithmetic hypothesis is imposed.

Theorem 1 (All vertex types)**.** *A finite-volume right-angled hyperbolic polytope has dimension at most eight.*

Theorem 2 (Entirely ideal polytopes)**.** *An entirely ideal finite-volume right-angled hyperbolic polytope has dimension at most four.*

Finite-volume right-angled examples in dimension eight are recalled in [5]. The regular ideal right-angled 24-cell gives the four-dimensional example. Thus both bounds are sharp.

1.1 Outline of the proofs

Both arguments begin with the same geometric observation. Projecting a finite interior point to a facet of maximum unit-normal height gives an interior point of that facet and decreases every inherited height by a controlled factor. Repetition produces a low-dimensional face whose outward normals determine a small spherical code.

For Theorem 1, three projections in dimension nine give a six-face. An explicit cubic sum of tensor squares then bounds its number of facets by 26, contradicting the known minimum 27. For Theorem 2, one projection in dimension five gives an ideal four-dimensional facet with at most 24 facets. The minimum-facet theorem identifies it with the regular 24-cell, and its opposite wall normals contradict the strict height bound. The projection is proved in Section 2; the two contradictions follow in Sections 3 and 4. Figures 1 and 2 summarize the two reductions. All algebra needed for these deductions is displayed in the paper.

1.2 Published geometric inputs

We use the following three numerical statements with their indicated scopes.

Proposition 3 (Six-dimensional facet minimum)**.** *A six-dimensional finite-volume right-angled hyperbolic polytope has at least* 27 *facets.*

This is the consequence of Lemma 1 and Proposition 4 of Dufour [2] explicitly recalled in [1, Section 4]. It applies to every vertex type, rather than only entirely ideal polytopes. We need neither a classification nor an equality statement for 27 facets.

Proposition 4 (Musin)**.** *A finite subset of* $S^3 \subset \mathbb{R}^4$ *with pairwise products at most* $1/2$ *between distinct members has cardinality at most* 24*.*

This is the Main theorem of [4, p. 4]. We use its cardinality bound, not uniqueness of a kissing configuration.

Proposition 5 (Kolpakov)**.** *An entirely ideal right-angled hyperbolic four-polytope has at least* 24 *facets. Equality holds only for the regular ideal right-angled* 24*-cell, up to hyperbolic isometry.*

This is Theorem 4 of the published paper [3, p. 1945]. The last two propositions are needed only for Theorem 2. The spherical bound used for Theorem 1 is proved directly in Section 3.

We use standard geometric foundations for acute-angled polytopes. In the Lorentz model, distinct outward unit facet normals have nonpositive products, including facets that do not share a ridge. As explained in [6, p. 42, §2.1], actual nonobtuse ridge angles imply the all-wall condition: nonadjacent supporting planes do not intersect, and the off-diagonal Gram entries are nonpositive. Faces of dimension at least three inherit right angles: every intrinsic ridge has a finite relative-interior point, where the incident walls are orthogonal. In the Klein closure, a face has vertices among those of the original polytope, so entire ideality is inherited. Triangulation into finite or ideal simplices gives finite volume. No projection chain below reaches a two-face, where ideal polygon angles would require separate treatment.

2 Descending maximum facet height

Use Lorentz signature $(-,+,\dots,+)$, future unit timelike points, and outward *unit* spacelike normals. Thus

$$P = \{x \in \mathbb{H}^n : \langle x, m\rangle \le 0 \text{ for every actual facet normal } m\}, \qquad \langle m, m'\rangle \le 0 \quad (m \ne m').$$

All heights below refer to this fixed unit-normal scaling. They are the hyperbolic sines of distances to the corresponding walls.

Lemma 6 (Farthest-facet projection)**.** *Let q be a finite relative-interior point of a right-angled polytope P, with $q^2 = -1$. Set $t(m) = -\langle q, m\rangle > 0$, choose a facet normal n with largest height $t = t(n)$, and put*

$$q' = \frac{q + tn}{\sqrt{1+t^2}}. \tag{1}$$

Then q' lies in the relative interior of the actual facet $Q = P \cap n^\perp$. Every intrinsic facet normal m_i of Q is an actual ridge neighbor of n, remains unit, and has height

$$0 < t_i' = -\langle q', m_i\rangle = \frac{t(m_i)}{\sqrt{1+t^2}} \le \frac{t}{\sqrt{1+t^2}} < 1. \tag{2}$$

Proof. The numerator in (1) has square $-1-t^2$ and product zero with n. Its product with q is $-1-t^2 < 0$, so normalization gives a future unit point in $n^\perp$. For *any* other actual facet normal m,

$$\langle q', m\rangle = \frac{\langle q, m\rangle + t\langle n, m\rangle}{\sqrt{1+t^2}} < 0. \tag{3}$$

The first numerator term is strict and the second is nonpositive. Thus q' lies in the actual facet and is strict against all other walls; it is not merely a formal point on the supporting plane.

An intrinsic facet of Q comes from its unique ridge neighbor m_i. Right angles give $\langle n, m_i\rangle = 0$, so restriction to $n^\perp$ preserves its unit norm. Formula (2) now follows from the maximum choice of t. □

If all heights at q are at most A, monotonicity of $x/\sqrt{1+x^2}$ gives a new bound $A/\sqrt{1+A^2}$. Start with the finite maximum A_0 at any finite interior point, and let A_j be the maximum height after j projections. Repeating the construction intrinsically gives

$$A_j \le \frac{A_0}{\sqrt{1+jA_0^2}} < \frac{1}{\sqrt{j}} \qquad (j \ge 1), \tag{4}$$

whenever the selected faces still have dimension at least three. The formula follows by induction, or by adding one to the inverse square of the previous bound. Strictness holds even if the maximizing facet is not unique at a step.

We will use the following translation from heights to spherical codes. At a unit point q of a d-dimensional face, choose q as its time axis and write the intrinsic unit normals as

$$m_i = \big(t_i, \sqrt{1+t_i^2}\, u_i\big), \qquad u_i \in S^{d-1} \subset \mathbb{R}^d. \tag{5}$$

For distinct facets, all-wall acuteness gives

$$u_i \cdot u_j \le \frac{t_i t_j}{\sqrt{(1+t_i^2)(1+t_j^2)}}. \tag{6}$$

The directions are distinct: if $u_i = u_j$, the Lorentz product of the two unit normals would be positive. Consequently the number of code points is exactly the number of facets of the selected face.

3 A strict spherical bound in six variables

Lemma 7. *A finite set of unit vectors in $\mathbb{R}^6$ whose distinct-pair products are strictly less than $1/4$ has at most 26 members.*

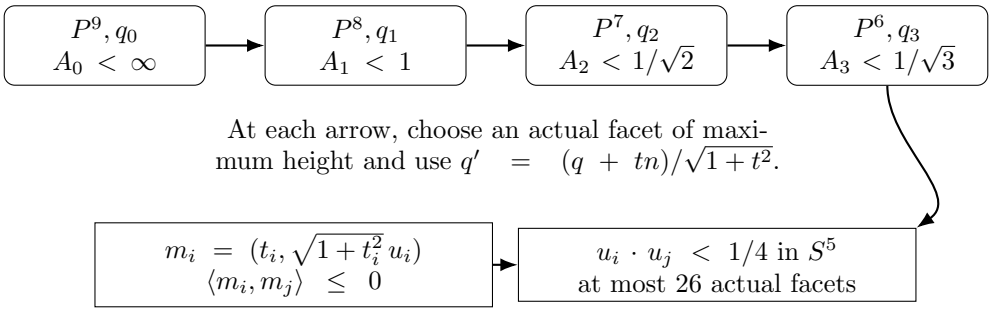


Figure 1: Three actual facet choices and the resulting spherical code. The arrows are the exact projection formulas, not ordinary edges of the polytope. The final code has one direction for each actual facet; the tensor calculation below contradicts the published minimum of 27.

Proof. Consider the cubic polynomial

$$F(x) = (x + \tfrac{1}{2})^2(x - \tfrac{1}{4}) = x^3 + \tfrac{3}{4}x^2 - \tfrac{1}{16}. \tag{7}$$

We give a direct tensor-square proof of the required positive-kernel identity, so no spherical linear-programming theorem is needed.

Use ordinary Euclidean tensor inner products. For a unit vector u define the rank-two tensor

$$A(u) = u \otimes u - I/6$$

and the symmetric rank-three tensors

$$S(u)_{abc} = u_a\delta_{bc} + u_b\delta_{ac} + u_c\delta_{ab}, \qquad T(u) = u^{\otimes 3} - S(u)/8.$$

For $x = u \cdot v$, contraction gives

$$\langle A(u), A(v)\rangle = x^2 - \tfrac{1}{6}, \qquad \langle T(u), T(v)\rangle = x^3 - \tfrac{3}{8}x. \tag{8}$$

Each cross contraction of a cubic tensor with S is $3x$; the remaining contraction is

$$\langle S(u), S(v)\rangle = (3 \cdot 6 + 6)x = 24x.$$

The scalar identity therefore becomes

$$\begin{aligned} F(u \cdot v) = {} & \tfrac{1}{16} + \tfrac{3}{8}(u \cdot v) \\ & + \tfrac{3}{4}\langle A(u), A(v)\rangle + \langle T(u), T(v)\rangle. \end{aligned}$$

Thus, for $u_1, \ldots, u_N$,

$$\begin{aligned} \sum_{i,j} F(u_i \cdot u_j) &= \frac{N^2}{16} + \frac{3}{8}\|\sum_i u_i\|^2 \\ &\quad + \frac{3}{4}\|\sum_i A(u_i)\|^2 + \|\sum_i T(u_i)\|^2 \\ &\geq \frac{N^2}{16}. \end{aligned} \tag{9}$$

On the other hand, $F(1) = 27/16$, and $F(x) \leq 0$ for $x < 1/4$. The diagonal terms and (9) give

$$\frac{N^2}{16} \leq \sum_{i,j} F(u_i \cdot u_j) \leq \frac{27N}{16},$$

so $N \leq 27$. If $N = 27$, equality throughout forces each off-diagonal polynomial value to vanish. Since $u_i \cdot u_j < 1/4$ strictly, the only available root is $-1/2$. Four such vectors would already give

$$\|u_1 + u_2 + u_3 + u_4\|^2 = 4 + 12(-\tfrac{1}{2}) = -2,$$

which is impossible. Hence $N \leq 26$. □

Proof of Theorem 1. Suppose first that a right-angled nine-polytope exists. Three projections from a finite interior point produce an actual six-face R and a relative-interior point at which every facet height is strictly less than $1/\sqrt{3}$, by (4). The inherited face is right-angled and has finite volume. For its intrinsic normals, (6) now gives

$$u_i \cdot u_j < \frac{1}{4} \qquad (i \neq j)$$

in $S^5 \subset \mathbb{R}^6$. Lemma 7 gives at most 26 facets for R, contradicting Proposition 3. Every higher-dimensional polytope has a nine-face inheriting finite volume and right angles, giving the same contradiction. □

4 The ideal bound from one projection

We first record the small metric calculation needed at equality. In the standard Lorentz model of the regular ideal right-angled 24-cell, its outward unit normals are

$$(1, \epsilon e_i + \delta e_j), \qquad 1 \leq i < j \leq 4, \quad \epsilon, \delta \in \{1, -1\}, \tag{10}$$

and the center is $(1, 0)$.

Lemma 8. *No future unit timelike point of the regular ideal right-angled 24-cell has all its unit-normal heights strictly less than one. More precisely, if every height is at most one, the point is the center and all the heights equal one.*

Proof. The normals in (10) come in opposite spatial pairs $(1, r), (1, -r)$. Write the point as (a, w), with $a > 0$ and $a^2 - \|w\|^2 = 1$. The two corresponding heights are $a - r \cdot w$ and $a + r \cdot w$. If both are at most one, their sum gives $a \leq 1$. Future unit timelikeness gives $a \geq 1$. Thus $a = 1, w = 0$, and every height equals one. Strict inequalities on all heights are impossible. □

Proof of Theorem 2. Assume an entirely ideal right-angled five-polytope exists. A single farthest-facet projection from a finite interior point gives a four-dimensional facet Q at whose relative-interior point all heights satisfy $0 < t_i < 1$. It inherits finite volume, entire ideality and right angles. By (6), its spherical directions in S^3 have pairwise products strictly below $1/2$. Proposition 4 bounds its facet number by 24; Proposition 5 gives the reverse bound and identifies Q with the regular ideal 24-cell. Its metric equality case, not a uniqueness assertion for kissing configurations, is what is used here. Lemma 8 contradicts the strict heights $t_i < 1$.

There is no entirely ideal right-angled five-polytope. In any higher dimension a five-face would inherit all three hypotheses and give the same contradiction. The ideal right-angled 24-cell attains dimension four. □

Both proofs use actual facets and a finite point checked against every wall. Neither infers a polytope from a set of formal Gram vectors, nor infers finite volume from a few feasible vertices. The strict inequalities come from projecting a finite point; they require no limiting or generic-position argument. For the general bound the only numerical geometric input is the six-dimensional facet minimum. For the ideal bound the additional inputs are exactly Musin's cardinality theorem and Kolpakov's metric equality theorem. All remaining calculations are displayed above.

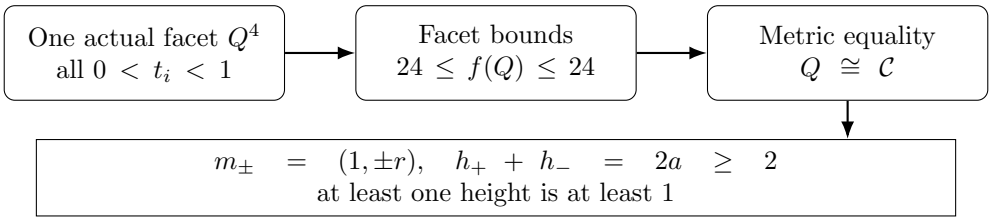


Figure 2: The ideal branch. Musin's spherical bound and Kolpakov's facet theorem force metric equality with the regular 24-cell. One opposite pair of its wall normals then contradicts the strict height bound produced by the projection. No uniqueness theorem for spherical kissing configurations is used.

5 Reproducibility and verification scope

The proofs above consist of geometric arguments and the displayed tensor identities, with the published inputs stated in Section 1. Accompanying exact Python checks, selected Lean 4 modules and their execution records are available in the repository snapshot `04c9bab7`. The catalogue entry for this paper is `right_angled_sharp`. Its two exact checks have the following precise purposes.

The coordinate check verifies the 24 normals in (10), their 276 unordered Gram products, the 12 opposite pairs and the displayed null vertices and cubical cusp links. This is a check of the specified model. The tensor check expands the 36 rank-two and 216 rank-three entries over the rational polynomial ring in twelve independent variables. It verifies (8) and the sum-of-squares identity (9), including the explicit remainders modulo the two unit-sphere equations. It also checks the ratio 27 and the strict endpoint calculation. These are exact identities; neither a numerical sample nor an enumeration of hyperbolic polytopes is involved.

From the repository root, inspect the selected jobs without starting a proof check:

```
python3 -B scripts/run_proof_batch_20261024.py manuscript \
  right_angled_sharp --mode scripts --plan
```

To execute the two checks and retain a receipt, choose a previously unused ledger filename:

```
python3 -B scripts/run_proof_batch_20261024.py manuscript \
  right_angled_sharp --mode scripts \
  --ledger results/sharp_replay_new.json
```

The exact checks require Python 3.11 or later. The guarded runner executes one job at a time with the repository's macOS resource monitor; the two checker sources use only the Python standard library. The optional `--mode core` selection additionally checks conditional Lean kernels for projection, the cubic bound and the dimension implications. The geometric realization, published theorems and face inheritance remain explicit premises of those kernels. Their scope is internal logic rather than a complete formalization of hyperbolic geometry.

The proof and reproduction guide identifies the checker sources and the mathematical statements they test. The release guide and source-bound verification record give the exact inputs, saved outcomes and separate formal-verification boundaries. These records concern their named source editions. The catalogue's PDF option builds its preserved publication edition; the present submission adds exposition and reproduction links while retaining the mathematical argument. Its `main.tex` is a standalone source: all figures are inline TikZ and the bibliography is included below.

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